MEMORANDUM

TO: Dr. Tina Han

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CC: ME32301 Lab Section 08- Harbin, cZachary; Alshaikh, Leen

DATE: February-21-2025

RE: Tensile Test to Failure Lab 2

EXECUTIVE SUMMARY

The purpose of this lab was to examine the linear or non-linear elastic response, transition to yielding, and ultimate failure for various materials, namely nylon, PVC, copper, steel, aluminum, and acrylic based on varying displacement rates. Furthermore, a primary objective of this lab was to understand if the material that failed in tension was ductile or brittle.

A virtual lab software Instron UTM 68SC-5 was used to perform the experiment on six materials: nylon, PVC, copper, steel, aluminum, and acrylic. These samples were selected with the size parameters detailed on Table 1, where L_t represents total length, L_0 represents gauge length, a_0 represents thickness, b_0 represents gauge width, and b_L represents the larger width of the sample.

Material	L_t[in]	L_0[in]	a_0[in]	b_0[in]	b_L [in]
Nylon	6	4	1/8	0.53	1
PVC	6	4	1/8	0.53	1
Copper	3	1.25	1/16	0.25	0.75
Steel	3	1.25	1/16	0.25	0.75
Aluminum	3	1.25	1/16	0.25	0.75
Acrylic	6	4	1/8	0.53	1

Table 1: Size parameters for the samples used in this lab.

The samples were then tested twice with the criteria detailed on Table 2, once with a slower displacement rate and then again with a faster displacement rate. $D_{r,s}$ represents the slow displacement rate, $D_{r,f}$ represents the fast displacement rate, D_{end} represents the end displacement criteria, F_{end}

represents the end force criteria, and L represents the travel limit.

Material	$D_{r,s}$[mm/min]	$D_{r,f}$[mm/min]	D_{end}[mm]	F_{end}[kN]	L[mm]
Nylon	2	20	200	4	381
PVC	2	20	200	4	254
Copper	0.1	1	50	4	127
Steel	0.1	1	50	4	127
Aluminum	0.1	1	50	4	127
Acrylic	0.1	1	200	4	254

Table 2: Displacement rates and end criteria of the samples.

After the data was collected, stress (σ) and strain (ε) values were calculated. Stress is the force applied divided by the cross-sectional area as seen in Equation (1). Strain is the change in length of the material divided by the original length as seen in Equation (2).

$$\sigma = \frac{F}{A_0} \tag{1}$$

$$\epsilon = \frac{\Delta L}{L_0} \tag{2}$$

Material	Acrylic	Aluminum	Nylon	PVC	Steel	Copper
Measured [GPa]	3.09	74.88	0.67	1.65	233.46	182.03
Published [GPa]	3.2	69	2	2.4	200	117
% Error	3.4	8.5	66.5	31.3	16.7	55.6

Table 3: Young's Modulus Comparison

The stress-strain curves gives insight into the elastic behavior and Young's modulus of the material, which is the slope of the graph. It also shows the the yield strength and ultimate strength. These factors are instrumental in meeting the goal of classifying the materials as ductile or brittle. The Table 3 shows the Young's Modulus obtained by analyzing stress-strain curves, along with errors associated with them. High percentage errors could be based on the material properties, the virtual lab environment's set programming to give biased results, and the assumption that testing conditions were devoid of experimental sources of error.

RESULTS

Subsequently, key points and regions from the graphs were evaluated. The graphs for each material with the two displacement rates per material have been indicated as follows:

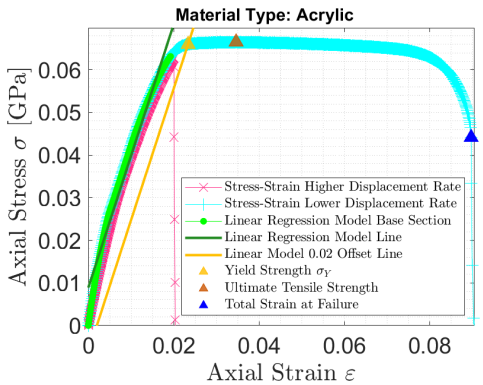

Figure 1: Acrylic

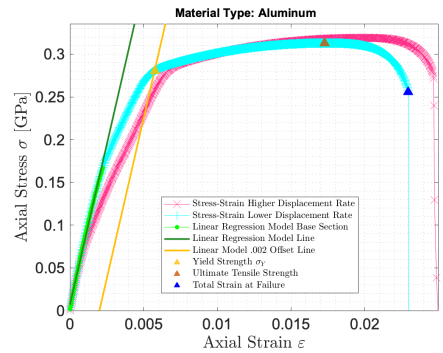

Figure 2: Aluminum

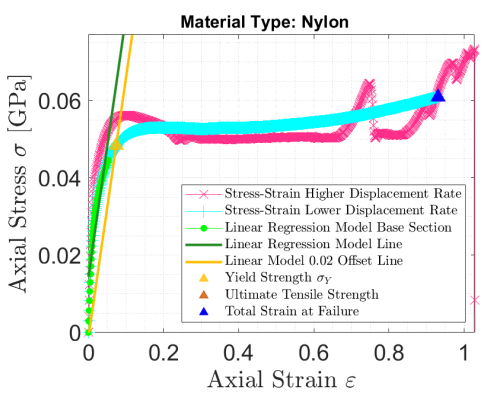

Figure 3: Nylon

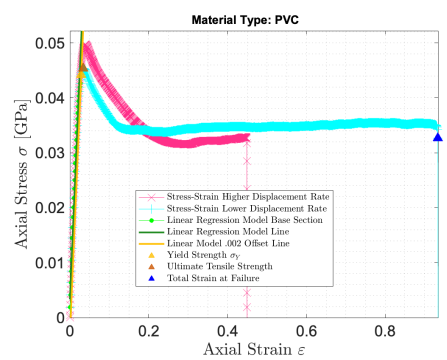

Figure 4: PVC

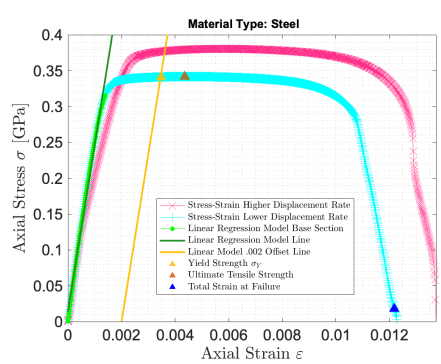

Figure 5: Steel

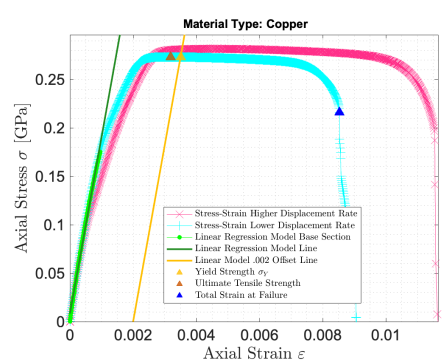

Figure 6: Copper

Figure 7: Stress-Strain Curves for Different Materials

The linear portion of the stress-strain curve is known as the elastic region. In this region, materials exhibit elastic behavior. The slope of the linear region, referred to as the Young's Modulus E, is a measure of a material's stiffness. Furthermore, the Yield Strength σ_Y of a material is the point where the material begins to undergo plastic deformation [1]. Since it is difficult to identify when a material changes from elastic to plastic deformation, the 0.2% offset method is used as can be seen in the yellow straight line on each graph. This method places a line parallel to the elastic region offset by 0.2% of strain [2]. The stress value corresponding to the point of intersection between the 0.2% offset line and the stress-strain curve is the Yield Strength of the material. In addition, the highest point on the stress-strain curve is known as the Ultimate Tensile Strength of a material σ_U. It is the maximum stress a material can endure before failure. The extent of a material's capacity to deform plastically without breaking is called Ductility. This property is quantified by the total strain at failure. The aforementioned material properties are shown in the stress-strain curves for each material in Figure 7. After analyzing these stress-strain curves, important parameters such as Young's Modulus, Yield Strength, and Ultimate Tensile Strength were calculated.

The experimental values of Young's modulus E from our data are compared to published values [3].

Material	Acrylic	Aluminum	Nylon	PVC	Steel	Copper
Measured [GPa]	3.09	74.88	0.67	1.65	233.46	182.03
Published [GPa]	3.2	69	2	2.4	200	117
% Error	3.4	8.5	66.5	31.3	16.7	55.6

Table 4: Young's Modulus Comparison

As can be seen in Table 4, the measured Young's modulus of acrylic is 3.4% lower than the published value, aluminum's is 8.5% higher, nylon's is 66.5% lower, PVC's is 31.3% lower, steel's is 16.7% higher, and copper's is 55.6% higher. This indicates significant error in Nylon and Copper's case, while Acrylic and Aluminum had fairly similar Young's Modulus based on published values.

The experimental values of yield strength σ_Y from our data are compared to published values [3] [4] [5].

Material	Acrylic	Aluminum	Nylon	PVC	Steel	Copper
Measured [MPa]	65.98	280.77	48.41	44.25	341.40	273.27
Published [MPa]	65	95	45	55	250	70
% Error	1.5	196	7.6	19.5	36.6	290.4

Table 5: Yield Strength Comparison

As can be seen in Table 5, the measured yield strength of acrylic is 1.5% higher than its published value, aluminum's is 196% higher, nylon's is 7.6% higher, PVC's is 19.5% lower, steel's is 36.6% higher, and copper's is 290.4% higher. Similar to Young's Modulus, Copper has a significantly high error, showing that this data collected was not as accurate in copper's case. However, unlike in the case of Young's Modulus, Aluminum has a high error in the case of Yield's strength. The other materials perform similar to their behavior in Young's Modulus case.

The experimental values of ultimate tensile strength σ_U from our data are compared to published values [3] [4] [6].

Material	Acrylic	Aluminum	Nylon	PVC	Steel	Copper
Measured [MPa]	66.54	313.14	60.91	45.41	341.91	273.40
Published [MPa]	70	110	60	37	400	220
% Error	4.9	184.7	1.5	22.7	14.5	24.3

Table 6: Ultimate Tensile Strength Comparison

As can be seen in Table 6, the measured ultimate tensile strength of acrylic is 4.9% lower than its published value, aluminum's is 184.7% higher, nylon's is 1.5% higher, PVC's is 22.7% higher, steel's is 14.5% lower, and copper's is 24.3% higher. Aluminum clearly showed a significant error as in the case of Yield Strength. The other materials showed low errors compared to previous performances.

Based on the discussion per table above and looking at the extent of plastic defomation in each material, PVC is the most ductile and copper is the least ductile. This classifications can be made by looking at how substantial the plastic region is on the stress-strain curves in Figure 7.

CONCLUSIONS

This lab successfully examines how nylon, PVC, copper, steel, aluminum, and acrylic behave under differing loading conditions, addressing their elastic response, as well as their failure.

Since this lab was virtual, there were not many sources of errors. However, it might not be a completely accurate simulation of reality, and there may be errors because of this. The main source could be a biased measurement fed into the program itself, that might not always take into consideration all random and absolute uncertainties in a real-world experimentation.

Based off the Results Section, (Table 4, Table 5, and Table 6) Acrylic has a Young's modulus of 3.09 GPa, meaning it elongates more under stress. Its yield strength is 66 MPa and its ultimate tensile strength is 66.5 MPa. Its plastic region is fairly flat but decreases slightly before dropping off at failure. Aluminum has a Young's modulus of 74.9 GPa, and it has a higher yield strength at around 281 MPa. Its yielding region increases meaning it experiences more axial stress for the same amount of strain, and its ultimate tensile strength is 313 MPa. Nylon has the lowest Young's modulus out of the tested materials at 0.67 GPa, meaning it is the flexible. Its yield strength is 48.4 MPa. Unlike the other materials, its yielding region does not dip down at higher strains close to failure, meaning the stress needed to elongate the material only increases. Its ultimate tensile strength is 60.9 MPa. It is the toughest material, as it has the greatest area under its stress-strain curve. PVC has a Young's modulus of 1.65 GPa, meaning it is fairly flexible. Its yield strength and ultimate tensile strength are close together at 44 MPa and 45 MPa respectively, and its yielding region dips initially, meaning it takes less stress to elongate the material, before flattening out. It is also the most ductile material because it has the highest total strain at 0.934. Steel has the highest Young's modulus out of the tested materials at 233 GPa, meaning it is the stiffest. Its yield strength and ultimate tensile strength are both around 341 MPa, the highest out of the tested materials, making it the strongest material. Its yielding region, though fairly flat, slowly dips down as it gets closer to failure. Copper also has a Young's modulus of 182 GPa. Like steel, its yield strength and ultimate tensile strength are around the same, about 273 MPa. The shape of its yielding

region is also similar to that of steel's, although its total strain at failure is less. It is the most brittle material because it has the smallest plastic region.

For materials like Nylon and PVC, the difference between high and low displacement rate curves is more noticeable, implying rate-dependent plastic deformation. In general, higher displacement rates tend to result in higher stress values at same strain levels. Interestingly, higher displacement rates tend to decrease ductility because the material has less time to undergo plastic deformation, causing lower strain at failure. Moreover, the material strength appears higher at higher displacement rates (golden triangles versus blue triangles in Figure 7, due to the strain hardening. This effect is more visible in ductile materials such as nylon and PVC where a higher strain rate increases resistance to deformation, reducing total strain but delaying failure.

It is important that the force limits of the machine are taken into consideration or else there could be heavy economic as well as physical damage to both the experimenter and the machine itself.

Lastly, after analyzing all the materials, each material can be fit into specific industries based on data from Table 4, Table 5, and Table 6. For instance, steel with the highest Young's Modulus (233GPa) and yield strength (341MPa) is ideal for construction and heavy machinery. Copper though strong (ultimate tensile strength of 274MPa), is brittle, and can be used in electrical applications. Aluminum (Young's Modulus of 74.9GPa, Ultimate Tensile Strength of 313MPa) strikes the balance between strength and flexibility making it suitable for aerospace and automotive applications. Nylon and PVC with low Young's Modulus as seen in Table 3, are suitable for insulating and damping applications. Lastly, as Acrylic is the most elastic (3.09GPa) and can be used in making transparent barriers.

Finally the lab's aim of understanding material properties based on strength, stiffness, and ductility parameters was successfully achieved.

REFERENCE LIST

[1] BYJUS. "Yield strength." (2022), [Online]. Available: https://byjus.com/physics/yield-strength/ (visited on 02/21/2025).

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[3] T. E. Toolbox. "Young's modulus, tensile strength and yield strength values for some materials." (2003), [Online]. Available: https://www.engineeringtoolbox.com/young-modulus-d_417.html (visited on 02/18/2025).

[4] MatWeb. "Overview of materials for acrylic, extruded." (), [Online]. Available: https://www.matweb.com/search/datasheet_print.aspx?matguid=632572aeef2a4224b5ac8fbd4f1b6f77 (visited on 02/18/2025).

[5] Varishapes. "Pvc." (2024), [Online]. Available: https://www.varishapes.com/materials-en/pvc (visited on 02/18/2025).

[6] MatWeb. "Overview of materials for pvc, rigid grade." (), [Online]. Available: https://www.matweb.com/search/datasheet_print.aspx?matguid=69642362cb864d25b8f6eb9d02092ecf (visited on 02/18/2025).

[7] OpenAI. "ChatGPT (February 21, 2025 Version)," [Online]. Available: https://openai.com/chatgpt (visited on 02/21/2025).

DATA SHEETS

The raw data collected in the lab using Instron UTM 68SC-5's Virtual Environment is presented in the following screenshots. As the data for Displacement (mm) and Force (kN) were very long and thus the first page was captured for each data set

Displacement (mm)	Force (kN)
0	-0.0002
0.0012	0.0027
0.009	0.0271
0.0184	0.056
0.0275	0.0831
0.0374	0.113
0.047	0.1411
0.0552	0.1648
0.064	0.1906
0.0745	0.2202
0.0835	0.2454
0.0936	0.2704
0.1025	0.2918
0.1114	0.3162
0.1201	0.3398
0.1302	0.3677
0.1394	0.392
0.1479	0.4139
0.1571	0.4377
0.1676	0.4645
0.1764	0.4866
0.186	0.5101
0.1954	0.533
0.2039	0.5537
0.2133	0.5758
0.2238	0.6009

Acrylic Data Sheet for 0.1 mm/min and 4kN-200mm

Displacement (m	Force (kN)
0	0.0005
0.0002	0.0008
0.0007	0.002
0.0019	0.0048
0.0028	0.0067
0.0048	0.0117
0.0075	0.0183
0.0094	0.0227
0.0113	0.0271
0.0134	0.0321
0.0155	0.0368
0.0173	0.041
0.0193	0.0457
0.0215	0.0505
0.0234	0.0549
0.0252	0.0588
0.0271	0.0629
0.0289	0.0671
0.0311	0.0717
0.0331	0.0764
0.0349	0.0802
0.0366	0.084
0.0385	0.0879
0.0407	0.0928
0.0428	0.0976
0.0447	0.1016

Acrylic Data Sheet for 1 mm/min and 4kN-200mm

Displacement (m	Force (kN)
0	-0.001
0.0007	0.0193
0.0014	0.0417
0.0022	0.0643
0.0029	0.0863
0.0036	0.1083
0.0044	0.1301
0.0051	0.1517
0.0058	0.1724
0.0065	0.193
0.0072	0.2089
0.008	0.2312
0.0087	0.2522
0.0095	0.2731
0.0102	0.2939
0.0109	0.3143
0.0116	0.3348
0.0124	0.3557
0.0131	0.3761
0.0138	0.3961
0.0145	0.416
0.0153	0.4365
0.016	0.4564
0.0167	0.4761
0.0175	0.4961
0.0182	0.5158

Aluminum Data for 0.1 mm/min and 4kN-50mm

Displacement (m	Force (kN)
0	0.0004
0.0001	0.0027
0.0007	0.0209
0.0015	0.045
0.0022	0.0683
0.003	0.0907
0.0038	0.1152
0.0046	0.1396
0.0053	0.1593
0.006	0.1799
0.0069	0.2037
0.0077	0.2249
0.0084	0.241
0.0093	0.2638
0.01	0.2841
0.0108	0.303
0.0116	0.3238
0.0124	0.3457
0.0131	0.363
0.0138	0.3799
0.0146	0.3991
0.0155	0.4193
0.0162	0.4368
0.017	0.4553
0.0178	0.4734
0.0185	0.4893

Aluminum Data Sheet for 1 mm/min and 4kN-50mm

Displacement (mm)	Force (kN)
0	0.0002
0.0909	0.1262
0.1851	0.2425
0.2794	0.3519
0.3732	0.4548
0.4683	0.5545
0.562	0.6465
0.6566	0.7337
0.7507	0.8143
0.8451	0.8888
0.9391	0.9561
1.0339	1.0179
1.1277	1.0718
1.2225	1.1211
1.3167	1.1639
1.4107	1.2011
1.5049	1.2347
1.5995	1.2666
1.6936	1.295
1.7881	1.3224
1.8825	1.3482
1.976	1.3713
2.0708	1.3951
2.1651	1.4176
2.2594	1.439
2.3538	1.46

Nylon Data Sheet for 2 mm/min and 4kN-200mm

Displacement (mm)	Force (kN)
0	0.0001
0.0893	0.1742
0.1938	0.3577
0.2984	0.5291
0.4021	0.6883
0.5064	0.8386
0.6101	0.9773
0.714	1.105
0.8187	1.2209
0.9228	1.3216
1.0272	1.4084
1.1313	1.4808
1.2357	1.5411
1.3391	1.5891
1.4435	1.6303
1.5474	1.6649
1.6514	1.695
1.7564	1.7241
1.8604	1.7493
1.9644	1.7727
2.0685	1.7956
2.1727	1.8176
2.2764	1.8376
2.3805	1.8576
2.4848	1.8779
2.5892	1.8976

Nylon Data Sheet for 20 mm/min and 4kN-200mm

Displacement (mm)	Force (kN)
0	-0.0038
0.0833	0.0826
0.1767	0.1771
0.2728	0.2717
0.3667	0.3618
0.4602	0.4494
0.5561	0.5371
0.6506	0.6208
0.7457	0.7028
0.8403	0.7818
0.9352	0.8584
1.0288	0.9308
1.1247	1.0031
1.2184	1.0707
1.314	1.1378
1.4083	1.2018
1.5016	1.2628
1.5977	1.3244
1.6919	1.3824
1.7867	1.4388
1.8814	1.493
1.9767	1.5453
2.0699	1.5933
2.1656	1.6407
2.2597	1.6838
2.3551	1.7246

PVC Data for 2 mm/min and 4kN-200mm

Displacement (mm)	Force (kN)
0	0.0001
0.0171	0.0185
0.0563	0.0603
0.0947	0.1007
0.1346	0.1417
0.1742	0.1819
0.2127	0.2204
0.2544	0.262
0.2951	0.3021
0.3341	0.3403
0.3736	0.3784
0.4137	0.4167
0.4553	0.4565
0.4947	0.4935
0.5343	0.5299
0.5746	0.5671
0.6146	0.6035
0.6551	0.6401
0.6947	0.6754
0.7341	0.7101
0.7743	0.745
0.8149	0.7801
0.8556	0.8144
0.8945	0.8465
0.9345	0.8791
0.9751	0.9115

PVC Data for 20 mm/min and 4kN-200mm

Displacement (mm)	Force (kN)
0	0.0005
0.0004	0.0358
0.0008	0.073
0.0012	0.1082
0.0016	0.1379
0.002	0.1741
0.0024	0.2095
0.0028	0.2448
0.0032	0.2811
0.0036	0.3167
0.004	0.3525
0.0044	0.388
0.0048	0.4235
0.0052	0.4592
0.0056	0.4945
0.006	0.5292
0.0064	0.5646
0.0068	0.5993
0.0072	0.6343
0.0076	0.6691
0.008	0.7036
0.0084	0.7382
0.0088	0.7725
0.0091	0.806
0.0096	0.8406
0.01	0.8746

Steel Data for 0.1 mm/min and 4kN-50mm

Displacement (mm)	Force (kN)
0	0.0002
0.0001	0.0067
0.0003	0.0191
0.0008	0.0357
0.0013	0.0554
0.0017	0.802
0.0021	0.1129
0.0026	0.1491
0.003	0.1826
0.0034	0.2166
0.0038	0.2583
0.0042	0.2943
0.0047	0.334
0.0051	0.3753
0.0056	0.4151
0.006	0.4501
0.0064	0.4866
0.0069	0.529
0.0072	0.5623
0.0077	0.6035
0.0081	0.6407
0.0085	0.6793
0.009	0.7145
0.0094	0.751
0.0098	0.788
0.0102	0.8185

Steel Data for 1 mm/min and 4kN-50mm

Displacement (mm)	Force (kN)
0	0.0007
0.0003	0.0227
0.0006	0.0434
0.0008	0.0652
0.0011	0.0868
0.0014	0.1083
0.0017	0.1294
0.002	0.1502
0.0023	0.1706
0.0025	0.1912
0.0028	0.2065
0.0031	0.2272
0.0034	0.2478
0.0037	0.2684
0.004	0.2887
0.0043	0.3086
0.0046	0.3283
0.0048	0.3482
0.0051	0.3684
0.0054	0.3883
0.0057	0.408
0.006	0.4275
0.0063	0.447
0.0065	0.4669
0.0068	0.4861
0.0071	0.5053

Copper Data for 0.1 mm/min and 4KN-50mm

Displacement (mm)	Force (kN)
0	-0.0002
0	0.0027
0.0004	0.0271
0.0007	0.056
0.0011	0.0831
0.0015	0.113
0.0019	0.1411
0.0022	0.1648
0.0026	0.1906
0.003	0.2202
0.0034	0.2454
0.0038	0.2704
0.0041	0.2918
0.0045	0.3162
0.0048	0.3398
0.0052	0.3677
0.0056	0.392
0.006	0.4139
0.0063	0.4377
0.0067	0.4645
0.0071	0.4866
0.0075	0.5101
0.0079	0.533
0.0082	0.5537
0.0086	0.5758
0.009	0.6009

Copper Data for 1 mm/min and 4KN-50mm

APPENDIX

This MATLAB script processes and analyzes experimental stress-strain data for a material tested at two different displacement rates. The script generates plots comparing stress-strain curves at different displacement rates, highlighting key material properties such as yield strength, ultimate tensile strength, and failure strain.

```
%%%
%You will need Statistics and Machine Learning Toolbox installed on Matlab
%to run this file. If you do not have the Toolbox installed, type "fitlm"
%in the Command Window, Matlab will promp you to install the Toolbox
%necessary.
%%%
clc          % Clean up Command Window
clear all    % Clean up all stored variables
close all    % Close all open windos

%% Preset Colors
% Do not change, this list includes 10 preset colors
% Feel free to use these colors in your plot
colorlist=["#6c3483","#CCCCFF", "#6495ED","#40E0D0","#9FE2BF",...
    "#DE3163","#FF7F50","#FFBF00","#f7dc6f","#2b7d38"];

%% Set up file names.
% Only Change the Filenames here to the file names you have without
% extension.
% For example, if matlab should read in test1slow.csv, Filenameslow should
% be set to 'test1slow'.
% This code process 1 material each time it is ran, so both tests with two
% different displacement rate should be imported.

Filenameslow="slowfilenamehere";
Filenamefast='fastfilenamehere';
```

```
Filetype='.csv';

%% Read Experimental Data
% Do not change, read the csv files to get displacement and force data from
% the two experiments of the same material
slow_data=readmatrix(append(Filenameslow,Filetype));
slow_disp=slow_data(:,1); %kN
slow_forc=slow_data(:,2); %mm

fast_data=readmatrix(append(Filenamefast,Filetype));
fast_disp=fast_data(:,1); %kN
fast_forc=fast_data(:,2); %mm


%% Input Sample Specs
% Edit to the correct Gauge Length, Thickness and Gauge Width. Make sure
% the units are converted to mm.

L0=101.6; % Gauge Length in mm
a0=3.175; % thickness in mm
b0=13.462; % Gauge Width in mm
A0=a0*b0; %mm^2

%% Calculate Stress and Strain data
% Do not change. Calculate stress and strain values based on the force and
% displacement readings

slow_stress=slow_forc./A0; %GPa
slow_strain=slow_disp./L0;
fast_stress=fast_forc./A0; %GPa
fast_strain=fast_disp./L0;

%% Generate Figure (1) to visualize two stress strain curves
```

```
% Do not change. This is to produce a figure comparing the two stress
% strain curves of the same material, continue to next section
figure (1)
plot(slow_strain,slow_stress,'--','LineWidth',2,'Color',colorlist(1));
hold on
plot(fast_strain,fast_stress,'--','LineWidth',2,'Color',colorlist(3));

%% Figure(1) properties
% Edit at need. This is just to change the axis labels and legend. However,
% this figure is not required for the memo submission. This figure is only
% for visualization purpose in the process.
box on
grid minor
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;

xlabel('Axial Strain $\varepsilon$','interpreter','latex');
ylabel('Axial Stress $\sigma$ [GPa]','interpreter','latex');

legend('Slow Experimental Data', ...
       'Fast Experimental Data', ...
       'FontSize', 11, 'Color', 'w'); % White background

%% Define which Set of data is used to calculate material properties
% Consider that there are two sets of stress and strain data, only one set
% should be used to estimate the material properties. Make the judgment of
% which set of data should be used and edit the following.

stress_data=slow_stress; % use either slow_stress or fast_stress
strain_data=slow_strain; % use either slow_strain or fast_strain

%% Automatically define the linear protion of the stress strain curve.
```

```
% This is based on function ischange(). Feel free to read the documentation
% online. Only thing that needs edit is the threshold value. Do not change
% anything else in this section. If the linear portion is not strict
% enough, lower the threshold value, if the linear portion can be wider,
% increase the threshold value.

% Adjust as needed
threshold=0.02; % recommended started value is 0.02

% use ischange() to find the linear portion, do not change
TF=ischange(stress_data,'linear','Threshold',threshold);

% Get index for the end of linear portion of the data, do not change
Linear_end=find(TF==1,1,'first');

% Define arrays for only linear portion for making the graph, do not change
strain_lin=strain_data(1:Linear_end);
stress_lin_data=stress_data(1:Linear_end);


%% Linear Regression Model of the data
% Dot not change the code in this section
% Use fitlm() to find Linear Regerssion Model (slope and intercept)
mdl=fitlm(strain_data(1:Linear_end),stress_data(1:Linear_end), ...
          'VarNames',{'Axial Stress','Axial Strain'});

% Export the intercept and slope value. Do not change
E_est=mdl.Coefficients.Estimate(2);
Int_est=mdl.Coefficients.Estimate(1);

% Calculate the data points for the linear regression model fo graph, do
% not change
stress_lin_fit=strain_data.*E_est+Int_est;
```

```

%% Find Yield Strength by the 0.2% offset method.
% The linear regression line is first shifted then compared to the
% experimental data value to find the interception. The stress value read
% of the interception is yield strength.

% Fill in the blank of the following line so that it creates a line with
% the same slope as and the linear regression model is intercepting with
% x-axis at 0.2% (equivalent to 0.002). Double check by looking at the
% figure(2) generated. stress_lin_shift should have the same length as
% strain_data and stress_data.
%%stress_lin_shift=strain_data*Int_est-0.002*E_est;
stress_lin_shift = strain_data * E_est - 0.002 * E_est;

% Compare the shifted linear fit with experimental data
check_yield=stress_lin_shift>=stress_data;

% Save the index of Yield strength
yield_index = find(check_yield, 1, 'first');

yield_ind = yield_index; % some code down the line references yield_ind

% Print the yield index
fprintf('Yield index: %d\n', yield_index);

%% Find Ultimate Tensile Strength
% Do not edit this section of the code
[ult_stress ult_ind]=max(stress_data);

%% Find Total Strain at Failure
% Edit this portion so that it automatically reports the strain at failure
% by comparing the each stress value to its previous reading after the
% linear elastic region. If the current stress value is less than 0.9 of
```

```
% the previous, the material fails.

% For example, if the stress reading in a row is [...;2;1.9;0.1;...], the
% material did not fail at 1.9 stress becuase 1.9>1.8. However, the material
% fails at 0.1 stress because 0.1<1.9*0.9.

% Before implementing this section, check the section of finding Yield
% Strength as it uses a very similar method.

% To implement this failure criteria, follow the listed steps.

% step 1: set up the criteria vector for the stress
% For the given example of stress [...; 4.0; 2.0; 0.5; 0.1; ...; ...],
% criteria vector should be       [...; ...; 3.6; 1.8; .45; .09; ...].
% Note that there is a shif in index.
% fail_crit should have the same length as stress_data.
fail_crit = [NaN; 0.9 * stress_data(1:end-1)]; % Shifted vector

% step 2: since the linear ealstic range is not considered for failure, set
% the linear portion of the fail_crit the same as the stress vector so the
% linear elatic region is never flagged as failure.
% Again, fail_crit should have the same length as stress_data.
fail_crit(1:Linear_end) = stress_data(1:Linear_end);

% step 3: check if the each measured stress meets the failure criteria.
% Refer to the comparison step in the yield strength section.
% check_fail should have the same length as stress_data, and should be 1
% when the corresponding stress meets the failure criteria and 0 for all
% other entries.
check_fail = stress_data < fail_crit;

% step 4: find the first index when the material is flagged for faliure.
% Refer to the Yield strength section code.
```

```
% Note that if the failure criteria is met, the previos value should be
% reported.
fail_ind = find(check_fail, 1, 'first') - 1;
fprintf("Fail index: %d\n", fail_ind)  % Print fail_ind as an integer

% step 5: in cases where the sample has a clear breaking point, stress
% readings might not meet the failure criteria implemented above.
% In those cases, use the last data point as the failure point.
if all(check_fail == 0)
    fail_ind = length(stress_data); % Use the last data point as the failure point
    fprintf("THIS MIGHT BE BAD?")
end


% step 6: Report strain at failure as variable strain_fail.
strain_fail = strain_data(fail_ind);

% If this section is set up correctly, figure(2) should reflect that total
% strain at failure is the strain right before a big drop in corresponding
% stress values.

%% Generating figure(2) to show how each material properties is estimated.
% The following section set up a graph by ploting each individual element.

% Initializing the figure.
% Do not change this following subsection of code
figure(2)
box on
grid minor
hold on

% Add a title at the top of the figure
```

```
sgtitle('Material Type: PVC', 'FontSize', 14, 'FontWeight', 'bold', 'FontName'
    , 'Arial');

% The following lines generate the plot of the following in this order:
% 1. Stress strain curve with the higher displacement rate.
% 2. Stress strain curve with the lower displacement rate.
% 3. Stress strain curve of the portion used for linear regression model.
% 4. Line generated based on the linear regression model.
% 5. .2% offset line from the linear regression model.
% 6. Yield strength.
% 7. Ultimate Tensile strength.
% 8. Total strain at Failure
% Do not change this following subsections of code.
fast_curve=plot(fast_strain,fast_stress);
slow_curve=plot(slow_strain,slow_stress);
lin_port=plot(strain_lin,stress_lin_data);
lin_fit=plot(strain_data,stress_lin_fit);
lin_shift=plot(strain_data,stress_lin_shift);
yield_plot=scatter(strain_data(yield_ind),stress_data(yield_ind));
ult_plot=scatter(strain_data(ult_ind),ult_stress);
fail_plot=scatter(strain_fail,stress_data(fail_ind));

% Edit as need, setup markers for stress strain curve with the higher
% displacement rate.
fast_curve.MarkerSize=10;
fast_curve.Marker='x';
fast_curve.Color='#FF2F89';
fast_curve.MarkerFaceColor='#FF6F50';

% Edit as need, setup markers for stress strain curve with the lower
% displacement rate.
slow_curve.MarkerSize=10;
slow_curve.Marker='+';
```

```
slow_curve.Color='c';
slow_curve.MarkerFaceColor='c';

% Edit as need, setup markers for stress strain curve of the portion used
% for linear regression model.
lin_port.MarkerSize=5;
lin_port.Marker='o';
lin_port.Color='g';
lin_port.MarkerFaceColor='g';

% Edit as need, setup markers for the line generated based on the linear
% regression model
lin_fit.LineStyle='-';
lin_fit.LineWidth=2;
lin_fit.Color='#228B22';

% Edit as need, setup markers for .2% offset line from the linear
% regression model.
lin_shift.LineStyle='-';
lin_shift.LineWidth=2;
lin_shift.Color='#FFBF00';

% Edit as needed, setup markers for yield strength.
yield_plot.SizeData = 100;
yield_plot.Marker = '^';
yield_plot.MarkerEdgeColor = '#FFBF00'; % Black outline
yield_plot.MarkerFaceColor = '#FFBF00'; % Black fill
yield_plot.MarkerFaceAlpha = 0.8;

% Edit as needed, setup markers for ultimate tensile strength.
ult_plot.SizeData = 100; % Recommended starting size 100
ult_plot.Marker = '^';
ult_plot.MarkerEdgeColor = "#CC5500"; % Orange outline
```

```
ult_plot.MarkerFaceColor = "#CC5500"; % Orange fill
ult_plot.MarkerFaceAlpha = 0.8;

% Edit as needed, setup markers for total strain at failure.
fail_plot.SizeData = 100; % Recommended starting size 100
fail_plot.Marker = '^';
fail_plot.MarkerEdgeColor = 'b'; % Blue
fail_plot.MarkerFaceColor = 'b'; % Blue
fail_plot.MarkerFaceAlpha = 1.0;

% Set up the axis limit so the range for shown stress is 0 to maximum
% measured stress. Do not change.
axis tight
ylim([0 max([fast_stress;slow_stress])*1.05]);

% Set up legend. Edit as needed. Make sure all legend is meaningful and the
% legend box does not block important information (set transparency). The
% following line also includes an example of how to use Latex editor to get
% Greek letter sigma.
legend('Stress-Strain Higher Displacement Rate','Stress-Strain Lower Displacement Rate', ...
    'Linear Regression Model Base Section','Linear Regression Model Line','Linear Model 0.02 Offset Line', ...
        'Yield Strength $\sigma_Y$','Ultimate Tensile Strength','Total Strain at Failure',...
        'FontName','Arial','FontSize',11, ...
        'Location','Southeast', ...
        'Color', 'w', ... % Set legend background to white
        'interpreter','latex');

% Set up axis labels. Edit as needed to include proper units etc.
xlabel('Axial Strain $\varepsilon$','interpreter','latex');
ylabel('Axial Stress $\sigma$ [GPa]','interpreter','latex');
```

```

% Set up axis font type and size. Edit as needed
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;


%% Export as PNG
% Edit as needed to export the generated graph as png file.

% To hide toolbar at the top right of the figure
set(figure(2), 'ToolBar', 'none');
% To export figure
exportgraphics(gcf,append(Filenameslow,'MaterialProperties.png'));

%% Print out material properties:
% Do not change this section.
fprintf('Estimated material properties based on chosen test are:\n');
fprintf('Young Modulus:             %9.5f [GPa]\n', E_est);
fprintf('Yield Strength:            %9.5f [MPa]\n', stress_data(yield_ind)
    *1000);
fprintf('Ultimate Tensile Strength: %9.5f [MPa]\n', ult_stress*1000);
fprintf('Total Strain at Failure:   %9.5f \n',strain_fail);
```